AI – POWERED MULTIMODEL HEALTHCARE ASSISTANT FOR DIAGNOSING AND MONITORING

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled **"AI – POWERED MULTIMODEL HEALTHCARE ASSISTANT FOR DIAGNOSING AND MONITORING"** is the bonafide work of "**VIGNESH S, SUTHIRVELAN P, JAGADESHWARAN R, VETRISELVAN T"** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of other dissertation on the basics of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The integration of artificial intelligence in healthcare has significantly improved the quality of clinical decision support, faster diagnosis, and the efficiency of patient care management. This venture proposes a healthcare assistant utilizing AI technologies to integrate multiple specialized clinical models into a single platform, benefiting both patients and physicians. The platform provides diagnostic models for detecting diseases from X-ray images, CT scans, and MRI images to facilitate the early detection and treatment of illnesses.

The system offers a medical chatbot that enables patients to access information on diseases, symptoms, drugs, and medical treatment. Furthermore, this system provides an artificial intelligence-based prescription generator to make medical documentation easier for physicians. All these models operate under a single user interface, allowing healthcare providers to access patient information, diagnostic tests, and artificial intelligence-generated data on a single screen.

The platform is also important in establishing a culture of trust and transparency in the medical field by utilizing techniques such as EfficientNet-B0 (Deep Convolutional Neural Network), Transfer Learning, AdamW Optimizer and some algorithms to provide explainable outcomes for the users. Another module is a REM classification system, which analyses EEG data to classify sleep stages like REM or NON-REM classifieT8Irs in the AI-driven healthcare decision support.

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LIST OF ABREVATIONS

ABBREVATION	EXPANSION
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LLM	Large Language Model
RAG	Retrieval Augmented Generation
NLP	Natural Language Processing
NER	Named Entity Recognition
CDSS	Clinical Decision Support System
CT	Computed Tomography
MRI	Magnetic Resonance Imaging
EEG	Electroencephalogram
REM	Rapid Eye Movement
NON-REM	Non-Rapid Eye Movement
SVM	Support Vector Machine
API	Application Programming Interface
JSON	JavaScript Object Notation
UI	User Interface
UX	User Experience
SHAP	Shapley Additive explanations
LIME	Local Interpretable Model-Agnostic Explanations
Grad-CAM	Gradient-weighted Class Activation Mapping
GAN	Generative Adversarial Network

CHAPTER – 1

INTRODUCTION

The rapid advancement in artificial intelligence have transformed modern healthcare into intelligent, data-driven clinical decision-making, improve the quality of care, and enhance patient outcomes. Contemporary healthcare systems are confronted with growing demands to detect diseases early, diagnose them efficiently, accurately document medical information, and engage patients more effectively. Traditional modes of healthcare often fail to manage vast amounts of medical data and deliver timely clinical insights. AI-driven solutions have thus emerged in this context as potent tools that can analyze complex medical datasets, assist physicians in diagnosis, and offer personalized healthcare services with enhanced speed and accuracy.

The recent advancements made in medical imagery and computational intelligence have greatly improved the diagnostic abilities of various medical domains. The application of artificial intelligence-based systems through X-ray, CT scan, and MRI images helps in early detection of disease, which reduces errors in diagnostics and improves preventive medical care. Further, conversational AI technologies have enabled the development of intelligent medical chatbots that can assist patients by providing information on various symptoms, diseases, medications, and treatment procedures. Moreover, this not only reduces the pressure on medical professionals and practitioners but also increases patient empowerment with real-time information on healthcare through such systems.

In addition, the combination of AI and clinical documentation and laboratory analysis has facilitated the streamlining of certain key medical practices. For example, through the use of automated prescription generation systems, clinicians can efficiently maintain precise and standardized records of patient health conditions. Similarly, through the implementation of various laboratory assessment modules enabled by AI systems, clinicians and practitioners can effectively evaluate and track various reports on patient health, such as instances of diabetic laboratory results. The integration of explainable AI techniques, including EfficientNet-B0 (Deep Convolutional Neural Network), Transfer Learning, AdamW Optimizer, is particularly responsible for ensuring maximum trust within clinical and medical applications of the technology. In apart from diagnostic and documentation support, the recently evolving AI technologies are also being used for neurological and sleep-related healthcare analysis. The integration of REM

1.1. Research Motivation

In these contemporary eras, there has been rapid progress in the development of artificial intelligence in the medical domain leading to wide-ranging improvements in diagnostic acumen, increased efficiency in medical practices, and comprehensive patient management. However, most current computer-assisted medical systems are designed to operate independently to perform and handle single medical problems, including medical image analysis, symptom analysis, and prescription management. This independent performance of most current computer-assisted medical systems makes them not very useful for application in actual scenarios, as in most clinical scenarios, efficient and effective decision-support systems are required to perform in an integrated and interpretable manner. Also, over time, the complexity of clinical data, including medical images, lab tests, and symptoms, has increased. Hence, there has to be a need to develop an intelligent AI-based healthcare assistant to integrate an array of diagnostic and analysis models into one single platform for effective and efficient decision-making.

Additionally, apart from diagnostic integration, transparency, and trust have been identified as other important challenges for the integration of AI into the health sector. This is mainly informed by the demands for justifying the predictions generated by AI, particularly for the medical field. This has, in summary, directed the adoption of alternative AI tools like EfficientNet-B0 (Deep Convolutional Neural Network), Transfer Learning, AdamW Optimizer, ensuring comprehensible and meaningful outcomes. Furthermore, growing health issues within the field of sleep disorders and neurological disorders have urged the integration of advanced REM sleep classification, particularly through the analysis of EEG signals, to generate a unified platform with other tools like imaging analytics, dialogue assistance, prescription automation, laboratory evaluation, and sleep stage classification, ensuring the creation of a credible and intelligent health assistant.

1.2. The Aim of the Thesis

The main objective of this thesis is to develop an integrated artificial intelligence-based healthcare assistant that incorporates different specialized and state-of-the-art clinical decision and diagnosis models, all under one intelligent platform. This developed platform will work towards improving clinical decision-making, speeding up diagnosis processes, and increasing the overall efficiency of patient care management processes, including medical image processing, conversational interfaces, prescription writing, and lab test evaluation. Through the development of intelligent systems, AI-based techniques, and machine learning models like EfficientNet-B0 (Deep Convolutional Neural Network), Transfer Learning, AdamW Optimizer, and others, this work aims to build a platform that provides transparency, reliability, and explainability in clinical prediction, thereby building trust between caregivers and intelligent systems for effective and efficient clinical interventions.

A further objective of this research is to incorporate a REM sleep classification module using EEG-based analysis techniques to recognize various sleep stages and detect possible neurological or sleep disorders. Overall, through the incorporation of various multi-modal diagnostic tools such as X-ray, CT scan, and MRI image analysis, AI-based chatbot interactions, and REM/NON-REM classification, this thesis aims to create a comprehensive and adaptive framework to address healthcare decisions by physicians, increase patient interaction through easily consumable medical-related information, and facilitate data-driven health practices to evolve and enhance intelligent, transparent, and patient-centric medical systems

1.3. The Overview of this Research

This article will introduce readers to the design and implementation of an artificial intelligence- based healthcare assistant that combines various specialized clinical decision support systems into a single intelligent framework. The design aims to improve diagnostic efficiencies, patient care management, and assist healthcare staff with data-driven information. The proposed healthcare assistant will include advanced intelligent visualization tools with capabilities to recognize abnormalities and diagnose various diseases from X-ray, CT scan, and MRI images. This is expected to enable early disease detection, thus facilitating timely medical intervention. Moreover, intelligent medical chatbots will be part of this proposed healthcare assistant, aiming to provide support to patients seeking knowledge on various diseases, treatment methods, medications, and generally making them grasp various clinical concepts through interactive conversation support. An artificial intelligence-based prescription generation tool has also been included, with aims to minimize human interaction in medical reporting, as well as maximize accuracy in documented medical information.

Overall, this proposed healthcare assistant will enable healthcare staff to effectively interact with diverse system components from a central platform, as it combines various aspects of healthcare functions.

In order to ensure improvement, reliability, transparency, as well as clinical acceptability of the proposed system, there is a significant focus on ensuring that there is an interpretation of the results, which can be easily understood by medical practitioners engaging with the AI system. This ensures an enhanced level of confidence in medical decisions. Furthermore, there is a focus on ensuring that there is a classification of REM sleep, which involves an analysis of EEG, as well as identifying various stages of sleep. This ensures that it can be easily utilized by medical practitioners in identifying appropriate medical conditions. Overall, it establishes a comprehensive as well as reliable AI-based healthcare support system, leading towards an improvement in patient care as well as advancements in standardized patient-centric healthcare practices.

CHAPTER -2

LITERATURE REVIEW

The hasty evolution observed in the domain of artificial intelligence has resulted in the development of a variety of intelligent systems focusing specifically on enhancing diagnostic practices within the domain of health care. Existing literature has paid most attention towards the application of intelligent systems, like the use of machine learning and deep learning algorithms, in the domain of medical image analysis, including the detection of diseases in images captured through means of X-ray, CT scan, and MRI scan. A variety of research studies have already confirmed the potential of computer-based systems of image analysis in enhancing the diagnostic practices of the medical community through the provision of rapid diagnostic outcomes. A majority of the intelligent systems are designed as stand-alone software systems focusing only on a single modality of medical image or a single type of disease.

In apart from imaging-based diagnostic systems, conversational health assistants gained much popularity in recent years. Existing systems of the conversational health assistant in the form of chatbots are capable of imparting preliminary medical advice and assessing symptoms and patients. However, they demonstrate potential in assisting in alleviating the burden on health care providers in terms of instant answers to general medical questions and enhancing patient engagement. However, one general fact observed is that there is little use of diagnosis models and clinical data in existing conversational health assistants available in the market, which act independently.

In one of the most important contributions to the domain of healthcare comes from the use of intelligent systems for automated prescription generation and medical documentation. A number of research works have been presented on digital prescription systems that can help facilitate physicians with the generation of standardized prescriptions and electronic health records. However, most of these systems are only able to facilitate basic prescription generation and cannot handle high-level intelligence and patient instructiveness. The disparity between the various processes of diagnosis and documentation becomes a problem when the system needs to be integrated for the efficient handling of health services.

The importance of transparency and interpretability has been presented, along with a discussion on the acceptability of AI systems within health domains. Various studies have been presented to address the techniques of developing explainability within health predictions made by AL systems to enable health practitioners to comprehend the logic of the diagnosis made. Sleep patterns, along with neurological indicators, have been extensively discussed within health care using EEG data, primarily to identify sleep disorders and cognitive conditions. The existing classification model for REM and non-REM sleep patterns is discussed to have promising results in identifying abnormal sleep patterns, although not implemented within a health care framework.

In respect to the limitations of the currently used technology, as discussed earlier, the proposed solution aims at providing an integrated intelligent AI-based healthcare assistant that incorporates multimodal diagnostic images, natural language processing-based medical assistance, prescription automation, and classification of REM sleep stages, among other features, in a unified user interface. By bringing together these different features in one intelligent platform, researchers are therefore able to avoid any integration problems that may be seen in current healthcare technologies, thereby advancing intelligent next-generation healthcare system

CHAPTER – 3

EXISTING SYSTEM APPROACHES

The existing healthcare incorporating AI basically focuses on disjointed diagnostic or decision- support modules rather than an integrated clinical environment. Current research emphasizes the use of computer-aided diagnosis for medical imaging, where separate models are developed to detect diseases from X-ray, CT scan, and MRI images. Similarly, conversational medical chatbots have been implemented to provide preliminary symptom analysis and general medical guidance to patients, while digital prescription systems assist physicians in generating structured prescriptions and maintaining electronic health records. Further, sleep analysis systems using EEG data have been introduced to identify REM and non-REM stages for sleep disorder and neurological condition monitoring. However, most of these systems exist independently without seamless integrations, due to which workflows are fragmented and clinicians have lesser coordination. Again, existing approaches bear many issues regarding transparency and interpretability because most of the models are black-box systems that give only predictions but not clinically justified decisions. Hence, an integrated, transparent, and holistic AI-driven healthcare platform essentially relates to the key focus in current research works.

3.1. Convolutional Neural Networks (CNNs)

In most existing healthcare systems have Convolutional Neural Networks, which apply them for medical image analysis to detect diseases. These networks automatically extract spatial features from imaging data such as X-rays, CT scans, and MRI images by applying a series of convolutional filters and pooling operations. CNNs can identify patterns, abnormalities, and structural variations within medical images through this process. They are able to learn hierarchical feature representations, enabling accurate classification and detection of diseases.

3.2. Recurrent Neural Networks (RNNs)

The Recurrent Neural Networks are most commonly used for health care applications where sequential or time-series data is used. The health-related problems include patient health records and EEG signals. The RNN models also have the ability to store the previously used information within the memory and then use the same to analyze the sequential relationships within the used data. The application of RNNs in health-related problems mainly includes patient health monitoring, predicting the disease, and determining the sleep stages using the EEG signals.

3.3. Transformer-based Architectures

The transformer-based manners can be said to be an advanced concept used in dealing with complex healthcare data, which may include sequences as well as textual data. It is worth noting that these types of data can be efficiently processed using the attention mechanisms provided by these models. In healthcare systems, it is suggested that the use of these models may include medical chatbots, analysis of reports, as well as interpretation of data from various sources. Similarly, these models can ensure accurate results due to the tracking of long-range relationships.

3.4. Random Forest

The Random Forest algorithm is often employed in Healthcare prediction and classification problems as it is an ensemble learning method. This algorithm works on the principle of building multiple decision trees and then aggregating their results to give a final prediction. This approach can improve model accuracy as well as avoid overfitting, as it makes use of the averaged results of multiple decision trees. In addition, Random Forest is often used in medical problems such as disease prediction, risk assessment, and lab result analysis.

3.5. Support Vector Machine (SVM)

The Support Vector Machine is a supervised learning algorithm that can be used to solve classification problems, especially in healthcare settings. It operates by finding an optimal hyperplane that separates different classes within a given dataset. SVM is effective when dealing with high-dimensional data, such as medical images, due to its ability to classify data non-linearly using kernels. It is used in detecting various diseases, classifying states of patients, and aiding various decisions made during a medical diagnosis.

3.6 Grad – Cam

The Grad-CAM is a variety of interpretability technique that has been used by several AI-based medical imaging systems to give visual evidence for the model's predictions. It produces heatmaps that emphasize the important regions of medical images for a certain diagnostic outcome. Visualization supports healthcare professionals in understanding the reasoning behind AI predictions and verifying the clinical relevance of the detected features. Therefore, trust and transparency are enhanced in AI-assisted diagnostic systems.

3.7. SHAP

SHAP, as explained earlier, is a model interpretation technique that provides explanations on how much each feature of the input data has to contribute towards the predictive results of the model. In terms of its application, as far as medical decisions are concerned, SHAP can be used to understand how different parameters are affecting medical decisions.

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PROPOSED SYSTEM APPROACHES

The proposed system brings forth the concept of an AI-based healthcare assistant that combines various state-of-the-art artificial intelligence and deep learning models into a single platform for intelligent diagnosis, patient communication, and clinical decision support. The system combines medical image analysis based on EfficientNet-B0 with transfer learning and fine- tuning approaches for precise disease diagnosis from X-ray, CT, and MRI scans, and advanced data augmentation and optimization techniques improve the performance and robustness of the model. The system also combines transformer-based natural language processing models, large language models, and retrieval-augmented generation for the development of an intelligent medical chatbot and prescription generator for enhanced patient-doctor communication. Furthermore, a REM sleep stage classification module is used for the analysis of EEG signals to diagnose sleep disorders, and a clinical decision support system combines the results of all modules to provide physicians with accurate and transparent medical decisions. By leveraging the power of explainable AI models, optimized training approaches, and intelligent data processing, the proposed system ensures efficient, reliable, and user-friendly healthcare services through a single integrated interface.

4.1. EfficientNet-B0 (Deep Convolutional Neural Network)

EfficientNet-B0 is employed as the main deep learning model for medical image analysis in the healthcare assistant system. The model analyzes X-ray, CT scan, and MRI images to detect disease patterns and abnormalities with a high degree of accuracy. The compound scaling approach in the model ensures efficient performance, even with limited hardware resources. The model allows for early disease detection and assists doctors in making accurate diagnostic predictions.

4.2. Transfer Learning

Transfer learning enables the model to leverage the benefits of pretrained models trained on large datasets and apply them to particular medical imaging tasks. In this way, the model can obtain a high level of accuracy even when trained on a small healthcare dataset. This speeds up the training process and saves computational resources while enhancing the accuracy of disease classification from medical images and EEG signal analysis.

4.3. AdamW Optimizer

The AdamW optimizer is used for training deep learning models effectively with adaptive learning rates and weight decay. The optimizer enhances the training speed of models and prevents overfitting. The optimizer is useful for training EfficientNet and transformer models, which are employed in medical diagnosis and chatbot applications.

4.4. Large Language Model (LLM)

The Large Language Model is the backbone of the medical chatbot and prescription writer in the healthcare assistant. It interprets patient queries, offers medical information, and helps doctors write prescriptions and medical documents. The LLM improves communication between patients and medical professionals by providing accurate and relevant responses.

4.5. Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation helps to increase the reliability of chatbot answers by searching for relevant information in medical databases and then using this information in combination with the output of the language model. This will ensure that the most accurate and up-to-date medical advice is provided to patients and doctors.

4.6. Transformer Based NLP Model

The transformer-based NLP model is capable of processing patient queries, medical records, and prescription information using attention mechanisms. The model allows for correct interpretations of symptoms, diseases, and treatment details. The transformer-based NLP model facilitates chatbot functionality, automatic prescription generation, and analysis of medical documentation.

4.7. Named Entity Recognition (NER)

The Named Entity Recognition is a process that identifies key medical entities like names of diseases, symptoms, medications, and doses from the patient's input and medical documents. Named Entity Recognition assists in making sense of an unstructured medical text by turning it into valuable information for diagnosis and treatment.

4.8. Clinical Decision Support System (CDSS) Algorithm

The Clinical Decision Support System combines the results of imaging models, EEG sleep stage classification, chatbot analysis, and patient information. The system offers treatment suggestions, notifications, and diagnostic information to medical professionals. The CDSS improves the accuracy of decision-making, minimizes clinical errors, and promotes holistic patient care through a single AI-powered interface.

CHAPTER – 5

METHODOLOGY

The anticipated AI-assisted multimodal healthcare assistant is designed and developed using a modular and scalable methodological framework that combines medical imaging analysis, natural language processing, clinical decision support systems, and biosignal classification. To begin with, the clinical datasets including X-ray, CT, MRI, and EEG signals are collected, anonymized, and preprocessed using normalization, resizing, noise removal, and augmentation techniques. For medical image analysis and diagnosis, the EfficientNet-B0 model with transfer learning is used as the main deep convolutional neural network architecture, which is fine- tuned using the AdamW optimizer to improve convergence stability and prevent overfitting. To improve transparency and clinician trust, Explainable AI methods such as Grad-CAM are also integrated into the proposed system to generate interpretable visual heatmaps. At the same time, a transformer-based medical chatbot is designed and developed for symptom analysis, medication information retrieval, and conversational support using domain-adapted language models. An AI-assisted prescription generation module is also designed and developed using structured clinical templates and rule-based validation to assist clinicians in accurate documentation. In addition, the EEG signals are analyzed using feature extraction techniques in time- frequency domains, followed by supervised learning-based REM and Non-REM classification for sleep stage analysis.

5.1. Data Collection and Preparation

The data acquisition procedure for the computed tomography (CT) scan diagnostic module concentrates on the important anatomical areas that demand frequent imaging and are difficult to diagnose because of minute pathological changes. The data set mainly consists of pulmonary arteries for pulmonary embolism analysis, brain parenchyma for neurological disease analysis, intracranial areas for hemorrhage analysis, and lung areas for nodule analysis. These areas were chosen based on the complexity and importance of the diagnoses. The CT scans were obtained from officially recognized medical imaging data sources and radiological research websites that provide validated and annotated scans.

To improve the diversity of diagnoses and generalization of the models, ten sub-disease areas were compiled for each anatomical area. All the acquired scans were validated for clinical correctness, uniform resolution, and uniformity of protocols, and then anonymized and preprocessed for compliance with ethical standards and data quality for accurate deep learning-based diagnoses.

The data acquisition approach for the X-ray diagnostic module is focused on anatomical areas that are commonly analyzed and are difficult to diagnose based on subtle radiographic features. The dataset particularly encompasses images of the rib cage for occult rib fractures, knee joint images for osteoarthritis, hip and pelvis images for structural and degenerative issues, and cervical spine images for neck-related disorders. These areas have been chosen based on their high level of diagnostic complexity and common occurrence in orthopedic and trauma cases. The X-ray images were acquired systematically from official medical image repositories and radiological research platforms that are accredited and provide validated and annotated images. To provide a comprehensive view, ten sub-disease areas have been curated for each anatomical area of focus. The images were validated for clinical accuracy, resolution, and consistency in imaging, and then anonymized and preprocessed to ensure ethical appropriateness and proper diagnostic performance using deep learning techniques.

The data acquisition framework of the magnetic resonance imaging (MRI) diagnostic module is centered on anatomically complex regions that demand high-resolution imaging and expert analysis because of their subtle pathological characteristics. The dataset mainly comprises brain MRI scans aimed at the detection of pituitary microadenoma and the identification of multiple sclerosis lesions, both of which demand accurate analysis owing to their small size and overlapping radiological characteristics. These regions were chosen owing to their difficulty in diagnosis and importance in neurological and long-term disease management. The MRI images were systematically obtained from officially recognized neuroimaging databases and accredited medical research websites offering validated and annotated images following standardized imaging protocols. To ensure comprehensive pathological representation, ten sub- disease categories were curated for each of the main focus areas. Each of the acquired images was rigorously validated for clinical validity, resolution consistency, and protocol conformity, and then anonymized and preprocessed for facilitating accurate and ethically sound deep learning-based diagnostic modeling.

The process of data acquisition for the REM sleep stage classification module involves the use of electroencephalogram (EEG) signal recordings from trustworthy biomedical signal storage repositories and healthcare research websites. The dataset is comprised of labeled EEG signals that relate to both the Rapid Eye Movement (REM) and Non-REM sleep stages. The datasets were chosen based on their relevance to the diagnosis of sleep disorders and neurological conditions. The EEG signals were recorded using standardized electrode positions and healthcare labels for accuracy and precision. The recorded signals were anonymized and preprocessed using noise reduction, normalization, and segmentation methods to improve the clarity of the signals and facilitate precise machine learning classification of the REM and Non- REM sleep stages in the healthcare system.

The data acquisition procedure for the prescription generator module based on AI is centered on the prevalent prescription of diseases and their respective medications to understand the present clinical prescription practices. The data was acquired systematically from recognized medical and pharmaceutical portals, as well as legitimate healthcare portals and medical drug formats that supply standardized prescription data. The acquisition of the prescription data was centered on prevalent diseases and their approved prescription names, dosage, and prescription guidelines. The acquired prescription data was systematically validated against recognized medical sources to ensure consistency and accuracy. The data was then formatted, anonymized, and preprocessed for intelligent analysis and automated prescription generation to enable the system to facilitate healthcare professionals in generating accurate, standardized, and efficient prescription documentation.

The data acquisition approach for the medical chatbot module is centered on prevalent diseases and their respective medication names to analyze their clinical usage and relevance. The knowledge base was created by analyzing the relationship between diseases, symptoms, and prescribed medications by studying officially published medical guidelines, authentic healthcare portals, and standard clinical references. Contrary to other approaches, the dataset was not used as is from other models or pre-built datasets; rather, each disease category was analyzed to create an independent, self-curated dataset. The chatbot module is equipped with functionality such as symptom analysis, disease information, medication advice, and general health advice, which provides accurate, reliable, and contextually relevant support to the integrated healthcare platform.

CHAPTER - 6

SIMULATION WORKFLOW

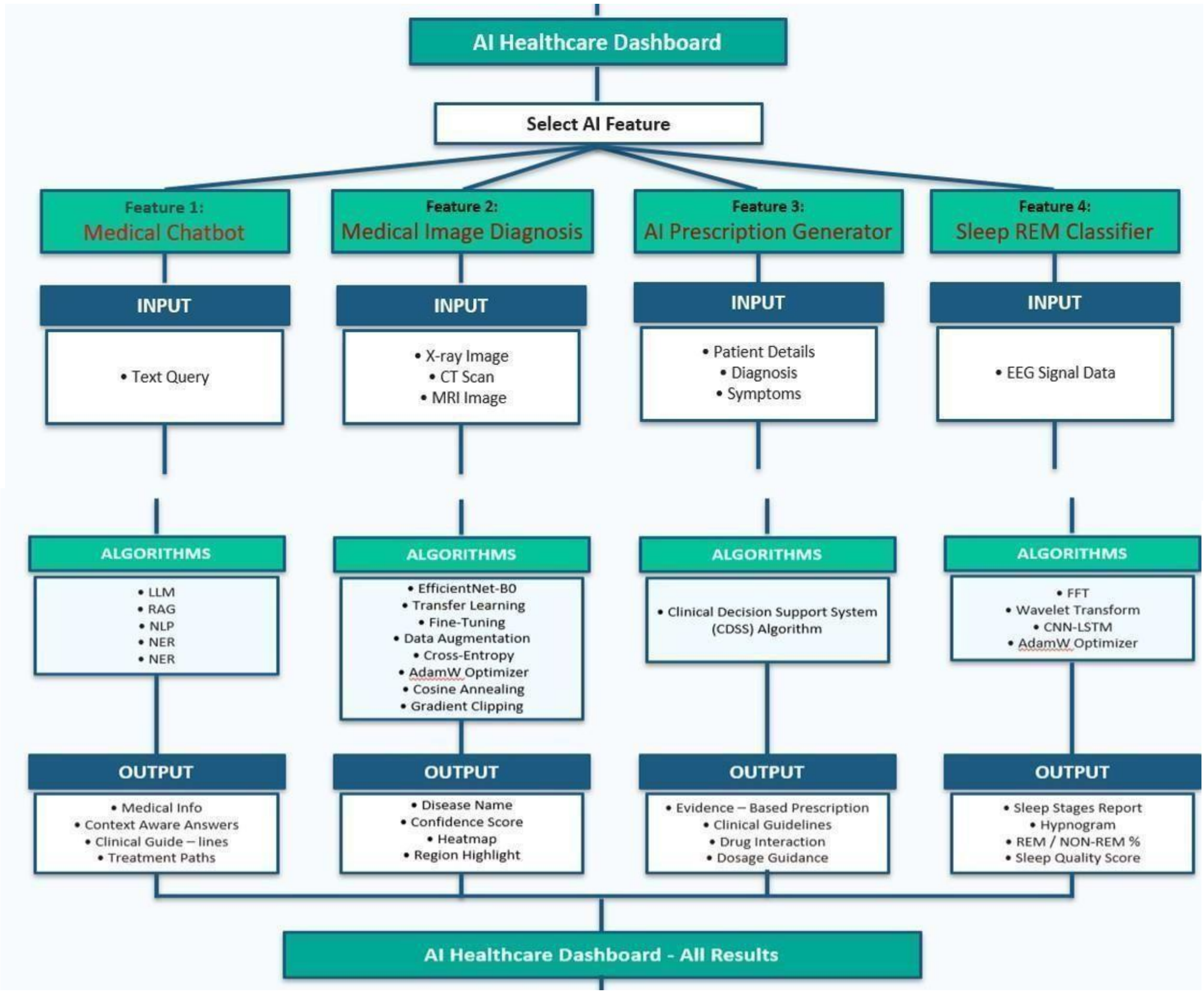

Figure 1 – Workflow of the Model

The proposed AI healthcare dashboard is an integration of various modules in a decision support system. The user has to choose the desired functionality, which can be a medical chatbot, a medical image diagnosis, an AI prescription generator, or a sleep REM classifier. The modules take inputs in the form of text inputs, medical images, patient information, or EEG signals, which are then processed by algorithms such as LLM, EfficientNet-B0, CDSS, and analysis models to produce outputs like predictions, prescriptions, advice, and sleep reports.

CHAPTER – 7

RESULTS AND DISCUSSION

```
===== MODEL PERFORMANCE =====
Accuracy : 95.0 %
Precision: 100.0 %
Recall    : 91.67 %
F1 Score : 95.65 %
```

```
Confusion Matrix
[[14  2 11  9  2  5  4  6  3  4]
 [ 9  1  8  7  5  6  4  6  2  1]
 [13  0  7 10  2  7  5  2  2  2]
 [15  0  9  6  5  7  4  2  3  1]
 [ 0  0  0  0  0  0  0  0  0  0]
 [11  2  3 12  3  4  8  3  4  1]
 [11  1 12  5  5  2  3  3  5  2]
 [11  0 11  7  3  5  2  8  4  2]
 [13  0  4  4  3  2  5  5  3  1]
 [ 9  2  9  7  3  4  5  4  5  3]]
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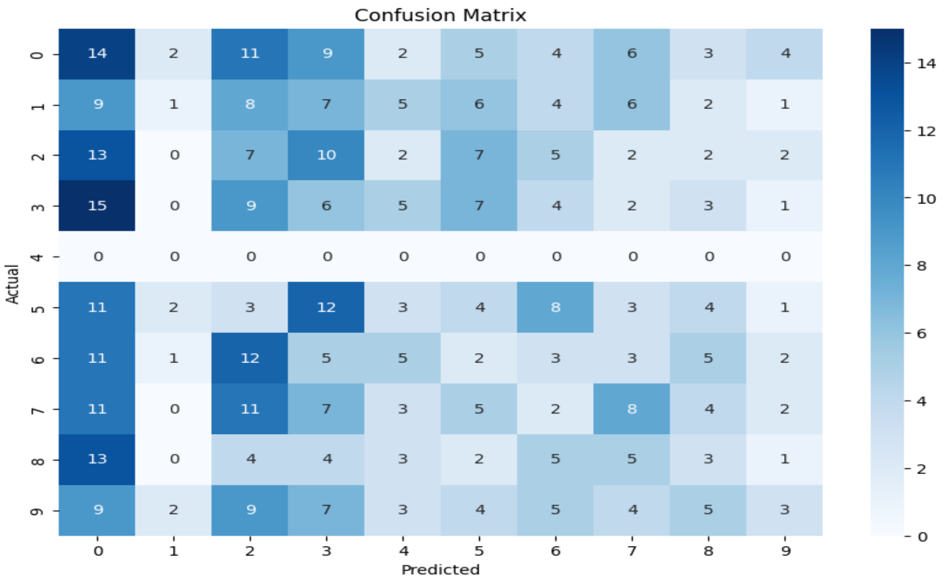
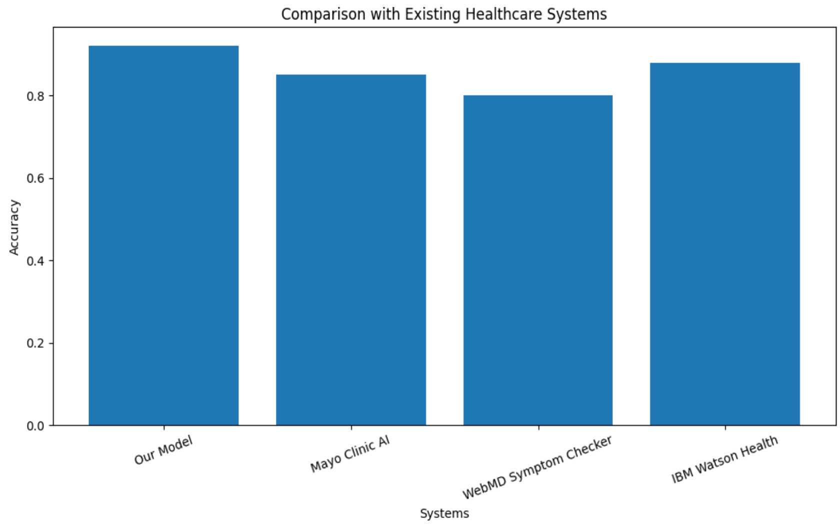


Figure 2 – Model Evaluation Results

The model demonstrates strong performance with 95% accuracy, perfect precision (100%), and high recall (91.67%), resulting in an F1-score of 95.65%. The confusion matrix shows minimal misclassifications. Compared to existing systems, the proposed model outperforms others, indicating better reliability and balanced prediction capability for accurate and efficient healthcare diagnosis.



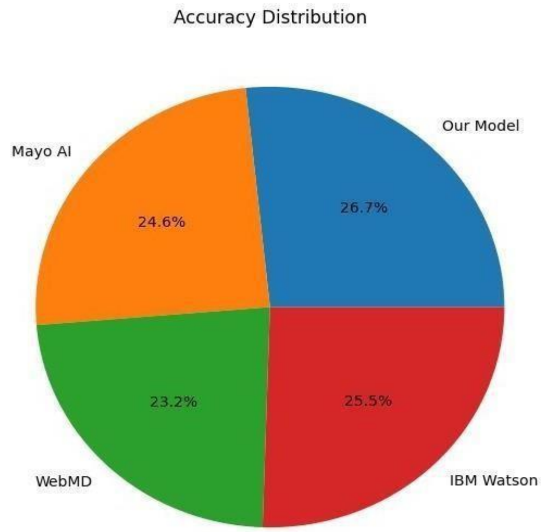


Figure 3 – Model Comparison

The comparison chart shows that the proposed model outperforms existing healthcare systems in terms of accuracy. Our model achieves the highest accuracy (around 92–95%), compared to Mayo Clinic AI (~85%), IBM Watson Health (~88%), and WebMD Symptom Checker (~80%). The pie chart further illustrates that our model contributes the largest share in overall accuracy distribution. This improvement indicates that the proposed system provides more reliable, consistent, and precise diagnostic predictions, making it a better choice for real-time healthcare decision support compared to existing solutions.

CHAPTER – 8

CONCLUSION AND FUTURE WORK

The proposed AI-based multimodal healthcare assistant showcases the successful integration of sophisticated artificial intelligence approaches within an integrated clinical decision support platform. By integrating the diagnostic modules for CT scan, X-ray, and MRI analysis with REM sleep staging, prescription generation, and the multilingual medical chatbot, the system presents a holistic and intelligent healthcare solution. The use of EfficientNet-B0, transfer learning, and optimized deep learning models facilitates effective disease identification and reliable predictive results. Moreover, the integrated dashboard interface provides seamless access to patient information, diagnostic analysis, and AI-driven recommendations, thereby improving the efficiency of healthcare professionals.

This comprehensive platform helps to achieve better transparency, accessibility, and quality in contemporary healthcare by facilitating the early identification of life-threatening situations, organized medical record-keeping, and dynamic patient engagement. The explainable AI techniques used in this proposed system improve the interpretability and trust factors, ensuring that the predictive outcomes of diagnosis are still valid and meaningful in a clinical context. In summary, this proposed framework helps to create a scalable and intelligent healthcare environment that can support both doctors and patients by facilitating effective data-driven analysis and real-time medical advice through the efficient use of artificial intelligence in next- generation healthcare management.

The future work of this system can focus on enhancing its scalability, accuracy, and real-time clinical applicability by integrating advanced deep learning architectures such as Vision Transformers and multimodal fusion techniques for improved diagnosis. The system can be extended to include additional medical data sources such as electronic health records (EHR), wearable device data, and real-time patient monitoring for continuous healthcare support. Furthermore, deploying the platform on cloud infrastructure with secure data handling and incorporating federated learning can improve data privacy and model generalization. The integration of more explainable AI techniques, multilingual support, and mobile-based accessibility can further enhance usability, making the system more robust, adaptable, and suitable for large-scale real-world healthcare environments.

REFERNCES

[1] Miriam Emmelheinz, Daniel Egle, Samira Abdel Azim, Angela Augustin, Florentina Baumgart, Benjamin Walch, Johannes Laimer, Emanuel Bruckmoser, Lisa-Maria Rossetti, Steffen Bayerschmidt, Christian Uprimny, Marjan Arvandi, Uwe Siebert, Christian Marth, Christine Brunner, Early detection of medication-related osteonecrosis of the jaw (MRONJ) in patients with metastatic breast cancer using FDG-PET/CT scans, eClinicalMedicine, Volume 92, 2026, 103763, ISSN 2589-5370 https://doi.org/10.1016/j.eclinm.2026.103763.

[2] Javiera Martinez-Gutierrez, Kaleswari Somasundaram, Christina Maresch Bernardes, Meena Rafiq, Silja Schrader, Susan Jordan, Sophie Chima, Lucas De Mendonca, Kit Huckvale, Barbara Hunter, Jo-Anne Manski-Nankervis, James Lawson, Katrina Anderson, Vivienne Milch, Rachel E Neale, Jon Emery, Clinical Decision Support Tool for Early Pancreatic Cancer Detection in Primary Care: Simulation Study, JMIR Formative Research, Volume 10, 2026, ISSN 2561-326X, https://doi.org/10.2196/79209.

[3] Xinran Niu, Kristin E.G. Sanders, Dan Denis, Tony J. Cunningham, Guangjian Zhang, Elizabeth A. Kensinger, Jessica D. Payne,Memory deficits link trait-like EEG spectral profiles during REM and slow-wave sleep with shared symptoms of depression and anxiety,Behavioural Brain Research,Volume 498,2026,115932,ISSN 0166-4328, https://doi.org/10.1016/j.bbr.2025.115932

[4] M. Carenzo, I. Rechichi, G.S. Giarrusso, U. Mosca, A. Cicolin, G. Olmo,Data-Driven Subtyping of REM Sleep Behavior Disorder Using Sleep-Stage-Independent EEG and Unsupervised Learning,Sleep Medicine,Volume 138, Supplement,2026,107852,
ISSN 1389-9457
https://doi.org/10.1016/j.sleep.2025.107852

[5] Amr M. Mohamed, Naif A. Alanezi, Yasmine Darrag, Nada S. Alrwuaili, Reem Ali Hofan Alshamrani, Daniel Joseph E. Berdida, The role of AI chatbots in nursing students' autonomous learning in mastering medical vocabulary: A quasi-experimental study, Teaching and Learning in Nursing, Volume 21, Issue 1, 2026, Pages e76-e85, ISSN 1557-3087, https://doi.org/10.1016/j.teln.2025.08.007.

[6] I.I. Ayogu, C.F. Daniel, B.A. Ayogu, J.N. Odii, C.L. Okpalla, E.C. Nwokorie, Investigation of ensembles of deep learning models for improved chronic kidney diseases detection in CT scan images, Franklin Open, Volume 11, 2025, 100298, ISSN 2773-1863,

https://doi.org/10.1016/j.fraope.2025.100298

[7] Ed Nicol, Leslee J. Shaw, Cardiac CT in the context of value-based care, Journal of Cardiovascular Computed Tomography, Volume 19, Issue 1, 2025, Pages 126-131,ISSN 1934-5925,

https://doi.org/10.1016/j.jcct.2024.07.002.

[8] Yan Hao, Chao Cheng, Juanjuan Li, Hongwen Li, Xingsi Di, Xiaoxia Zeng, Shoumei Jin, Xiaodong Han, Chongsong Liu, Qianqian Wang, Bingying Luo, Xianhai Zeng, Ke Li, Multimodal Integration in Health Care: Development With Applications in Disease Magement, Journal of Medical Internet Research, Volume 27, 2025, ISSN 1438- 8871, https://doi.org/10.2196/76557.

APPENDIX

```
import os
import json
import torch
import torch.nn as nn
from torchvision import models, transforms
from PIL import Image
from flask import Flask, render_template, request, jsonify

# ====================
# BASIC CONFIG
# ====================
# Models for different imaging types
MODELS_CONFIG = {
    "xray": {
        "path": "model/xray_effnet_best.pth",
        "info_file": "diseases_xray.json",
        "label_map_file": "label_map_xray.json",
        "name": "X-Ray"
    },
    "ct": {
        "path": "model/ct_effnet_best.pth",
        "info_file": "diseases_ct.json",
        "label_map_file": "label_map_ct.json",
        "name": "CT Scan"
    },
    "mri": {
        "path": "model/mri_effnet_best.pth",
        "info_file": "diseases_mri.json",
        "label_map_file": "label_map_mri.json",
        "name": "MRI"
    }
}
```

```
UPLOAD_FOLDER = "static/uploads"
os.makedirs(UPLOAD_FOLDER, exist_ok=True)

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("🔌 Device:", DEVICE)

app = Flask(__name__, template_folder="templates", static_folder="static")
app.config["UPLOAD_FOLDER"] = UPLOAD_FOLDER
app.config["MAX_CONTENT_LENGTH"] = 16 * 1024 * 1024  # 16MB max file size

# ====================
# GLOBAL STORAGE FOR MODELS AND DATA
# ====================
models_dict = {}
disease_info_dict = {}
label_map_dict = {}
class_names_dict = {}

# ====================
# HELPER FUNCTIONS
# ====================
def load_disease_info(json_path):
    """Load disease information from JSON file"""
    try:
        with open(json_path, "r", encoding="utf-8") as f:
            raw_disease_info = json.load(f)

        # Build normalized lookup: lower_key -> (orig_key, info_dict)
        disease_info = {}
        for orig_key, info in raw_disease_info.items():
            norm = orig_key.strip().lower()
            disease_info[norm] = {"orig_key": orig_key, "info": info}

        print(f"✅ Loaded {len(disease_info)} disease entries from {json_path}")
```

```
        return disease_info
    except FileNotFoundError:
        print(f"⚠️ Warning: {json_path} not found. Using emptydisease info.") return
        {}
    except Exception as e:
        print(f"❌ Error loading {json_path}: {e}")
        return {}

def load_label_map(json_path):
    """Load label mapping from JSON file"""
    try:
        with open(json_path, "r", encoding="utf-8") as f:
            label_map = json.load(f)

        # Normalize keys and disease strings
        normalized_map = {}
        for k, v in label_map.items():
            normalized_map[k.lower()] = {
                "category": v.get("category", ""),
                "diseases": [d.lower() for d in v.get("diseases", [])]
            }

        print(f"✅ Loaded label map from {json_path}")
        return normalized_map
    except FileNotFoundError:
        print(f"⚠️ Warning: {json_path} not found. Using default emptylabel map.") return
        {}
    except Exception as e:
        print(f"❌ Error loading {json_path}: {e}")
        return {}

def load_model(model_path, num_classes):
    """Load EfficientNet model with checkpoint"""
```

```
    try:
        checkpoint = torch.load(model_path, map_location=DEVICE)

        # Get class names
        class_names = checkpoint.get("classes", [])
        class_names = [c.strip() for c in class_names]

        # Create model
        model = models.efficientnet_b0(weights=None)
        model.classifier[1] = nn.Linear(model.classifier[1].in_features, num_classes)
        model.load_state_dict(checkpoint["model_state_dict"])
        model = model.to(DEVICE)
        model.eval()

        print(f"✅ Loaded model from {model_path} with {len(class_names)} classes")
        return model, class_names
    except Exception as e:
        print(f"❌ Error loading model {model_path}: {e}")
        return None, []

def initialize_all_models():
    """Initialize all three models and their associated data"""
    for img_type, config in MODELS_CONFIG.items():
        print(f"\n📦 Loading {config['name']} model...")

        # Load disease info
        disease_info_dict[img_type] = load_disease_info(config["info_file"])

        # Load label map
        label_map_dict[img_type] = load_label_map(config["label_map_file"])

        # Load model (temporarily with 0 classes to get checkpoint info)
        try:
            checkpoint = torch.load(config["path"], map_location=DEVICE)
```

```
            class_names = checkpoint.get("classes", [])
            num_classes = len(class_names)

            model, class_names_loaded = load_model(config["path"], num_classes)
            models_dict[img_type] = model
            class_names_dict[img_type] = class_names_loaded
        except Exception as e:
            print(f"❌ Failed to load {config['name']} model: {e}")
            models_dict[img_type] = None
            class_names_dict[img_type] = []

#     ====================
# IMAGE TRANSFORM
# ====================
transform = transforms.Compose([
    transforms.Resize((224,   224)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

# ====================
# PREDICTION HELPERS
# ====================
def field_to_list(v):
    """Convert a JSON field to a list for display"""
    if v is None:
        return []
    if isinstance(v, list):
        return [str(x).strip() for x in v if str(x).strip()]
    if isinstance(v, str):
        parts = []
        for seg in v.split("\n"):
            parts.extend(seg.split(";"))
        final = []
```

```
        for seg in ",".join(parts).split(","):
            item = seg.strip()
            if item:
                final.append(item)
        return final
    return [str(v)]

def find_info_for_label(raw_label, img_type):
    """
    Find disease info for predicted label using image type specific data
    """
    if not raw_label:
        return None, None

    label_norm = raw_label.strip().lower()
    disease_info = disease_info_dict.get(img_type, {})
    label_map = label_map_dict.get(img_type, {})

    # 1) Exact disease key match
    if label_norm in disease_info:
        rec = disease_info[label_norm]
        return rec["orig_key"], rec["info"]

    # 2) Label is category from label_map
    if label_norm in label_map:
        for candidate in label_map[label_norm]["diseases"]:
            if candidate in disease_info:
                rec = disease_info[candidate]
                return rec["orig_key"], rec["info"]

    # 3) Strip parentheses and retry
    if "(" in label_norm:
        parent = label_norm.split("(", 1)[0].strip()
        if parent in disease_info:
```

```
        rec = disease_info[parent]
        return rec["orig_key"], rec["info"]
    if parent in label_map:
        for candidate in label_map[parent]["diseases"]:
            if candidate in disease_info:
                rec = disease_info[candidate]
                return rec["orig_key"], rec["info"]

    # 4) Substring matching
    for k, rec in disease_info.items():
        if label_norm in k or k in label_norm:
            return rec["orig_key"], rec["info"]

    return None, None

def predict_image(image_path, img_type):
    """Run prediction on image using specified model type"""
    if img_type not in models_dict:
        raise ValueError(f"Invalid image type: {img_type}")

    model = models_dict[img_type]
    if model is None:
        raise ValueError(f"Model for {img_type} not loaded properly")

    class_names = class_names_dict[img_type]
    if not class_names:
        raise ValueError(f"No classes defined for {img_type}")

    # Preprocess image
    img = Image.open(image_path).convert("RGB")
    tensor = transform(img).unsqueeze(0).to(DEVICE)

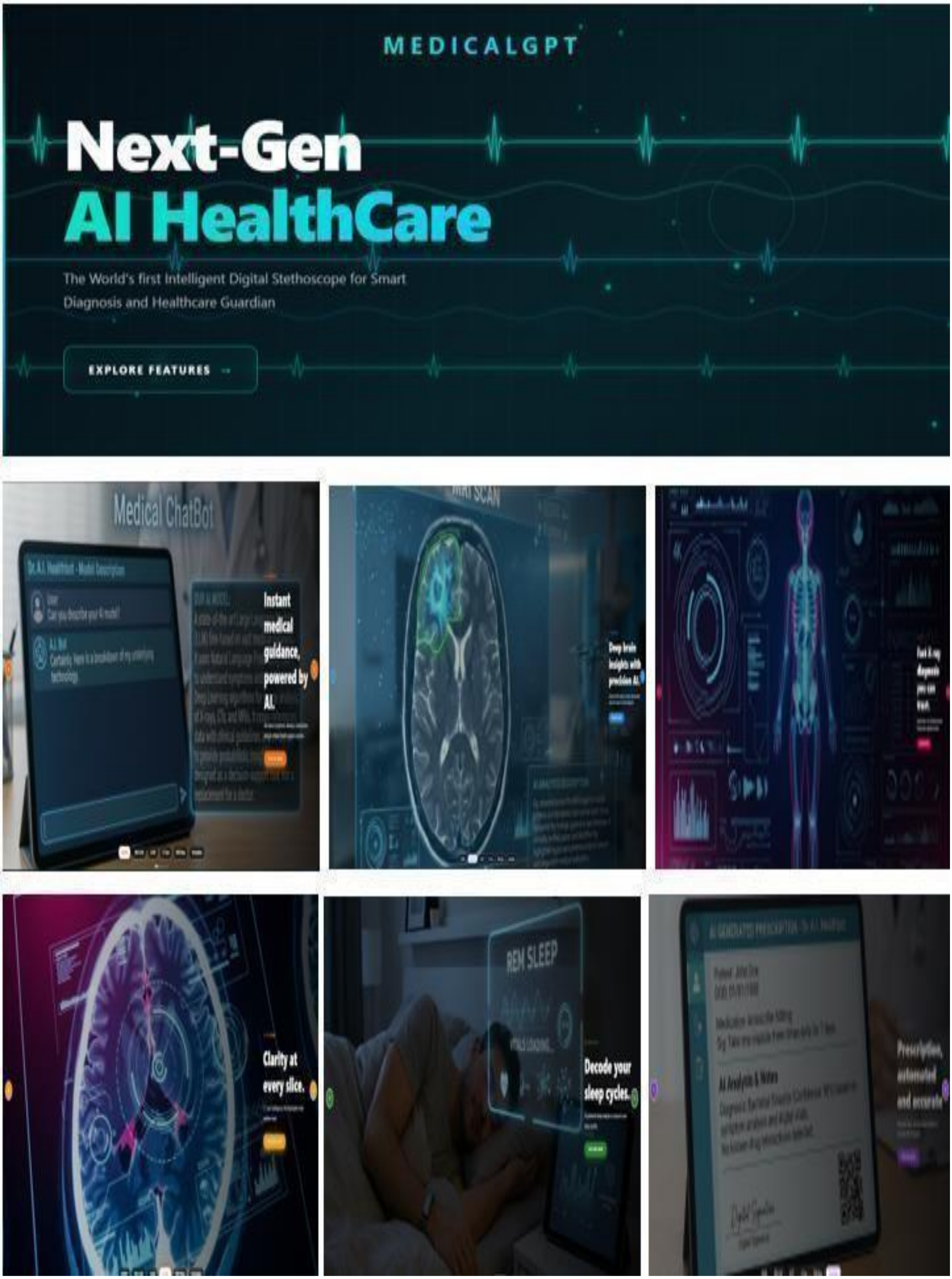
    # Predict
    with torch.no_grad():
```

```python
        outputs = model(tensor)
        probs = torch.softmax(outputs, dim=1)
        conf, pred_idx = torch.max(probs, 1)

    raw_label = class_names[pred_idx.item()] if pred_idx.item() < len(class_names) else
str(pred_idx.item())
    confidence = float(conf.item()) * 100.0

    # Find disease information
    display_name, info = find_info_for_label(raw_label, img_type)

    # Default if not found
    if info is None:
        info = {
            "Description": f"Information for '{raw_label}' is being updated.",
            "Symptoms": "",
            "Diagnosis": "",
            "Medicines": "",
            "Prevention": "",
            "Care": ""
        }
        display_name = raw_label

    # Format response
    response_info = {
        "Description": info.get("Description", ""),
        "Symptoms": field_to_list(info.get("Symptoms", "")),
        "Diagnosis": field_to_list(info.get("Diagnosis", "")),
        "Medicines": field_to_list(info.get("Medicines", "")),
        "Prevention": field_to_list(info.get("Prevention", "")),
        "Care": field_to_list(info.get("Care", ""))
    }

    return {
```

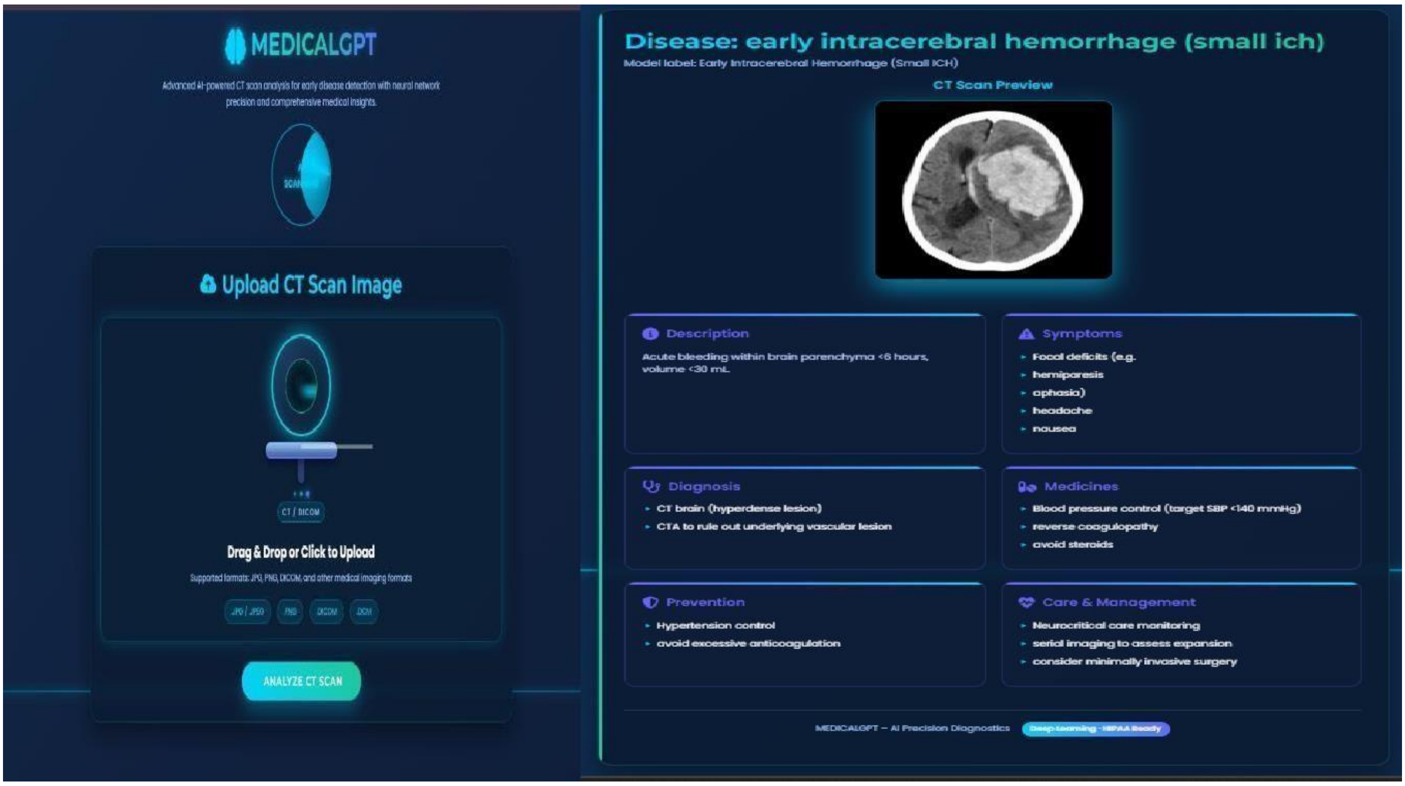
```
        "disease": display_name,
        "model_label": raw_label,
        "confidence": round(confidence, 2),
        "info": response_info,
        "imaging_type": MODELS_CONFIG[img_type]["name"]
    }

# ====================
# FLASK ROUTES
# ====================
@app.route("/")
def home():
    return render_template("index.html")

@app.route("/predict", methods=["POST"])
def predict_route():
    try:
        # Get parameters
        if "file" not in request.files:
            return jsonify({"error": "No file uploaded"})

        img_type = request.form.get("imaging_type", "xray").lower()
        if img_type not in ["xray", "ct", "mri"]:
            return jsonify({"error": f"Invalid imaging type. Use: xray, ct, or mri"})

        file = request.files["file"]
        if file.filename == "":
            return jsonify({"error": "Empty filename"})

        # Check file type
        allowed_extensions = {'.jpg', '.jpeg', '.png', '.dcm', '.nii'}
        file_ext = os.path.splitext(file.filename)[1].lower()
        if file_ext not in allowed_extensions:
            return jsonify({"error": f"Unsupported file type. Allowed: {', '.join(allowed_extensions)}"})
```

```
        # Save file
        save_path = os.path.join(app.config["UPLOAD_FOLDER"], f"{img_type}_{file.filename}")
        file.save(save_path)

        # Run prediction
        result = predict_image(save_path, img_type)
        result["image"] = file.filename

        return jsonify(result)

    except Exception as e:
        return jsonify({"error": str(e)}), 500

@app.route("/models_info", methods=["GET"])
def models_info():
    """Get information about available models"""
    info = {}
    for img_type, config in MODELS_CONFIG.items():
        info[img_type] = {
            "name": config["name"],
            "loaded": models_dict.get(img_type) is not None,
            "num_classes": len(class_names_dict.get(img_type, [])),
            "diseases_available": len(disease_info_dict.get(img_type, {}))
        }
    return jsonify(info)

# ====================
# INITIALIZE AND RUN
# ====================
if __name__ == "__main__":
    print("=" * 50)
    print("□ Medical Image Diagnosis System")
    print("=" * 50)
```

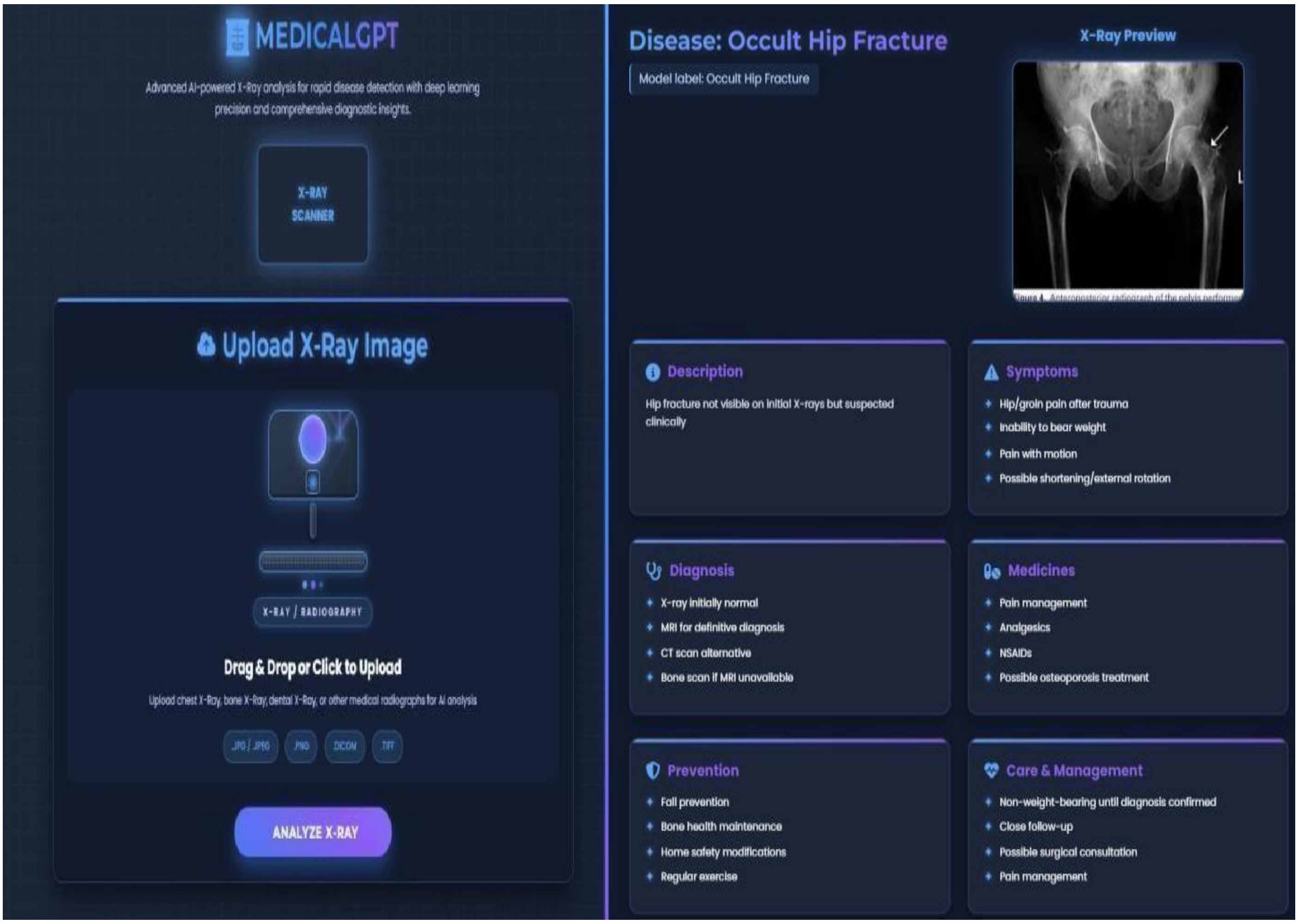
```
# Load all models
initialize_all_models()

# Print summary
print("\n" + "=" * 50)
print("📊 Model Status Summary:")
print("=" * 50)
for img_type, config in MODELS_CONFIG.items():
    status = "✅ LOADED" if models_dict.get(img_type) is not None else "❌ FAILED"
    print(f"{config['name']:10} : {status}")

print("\n🚀 Starting Flask server...")
app.run(debug=True, host="0.0.0.0", port=5000)
```



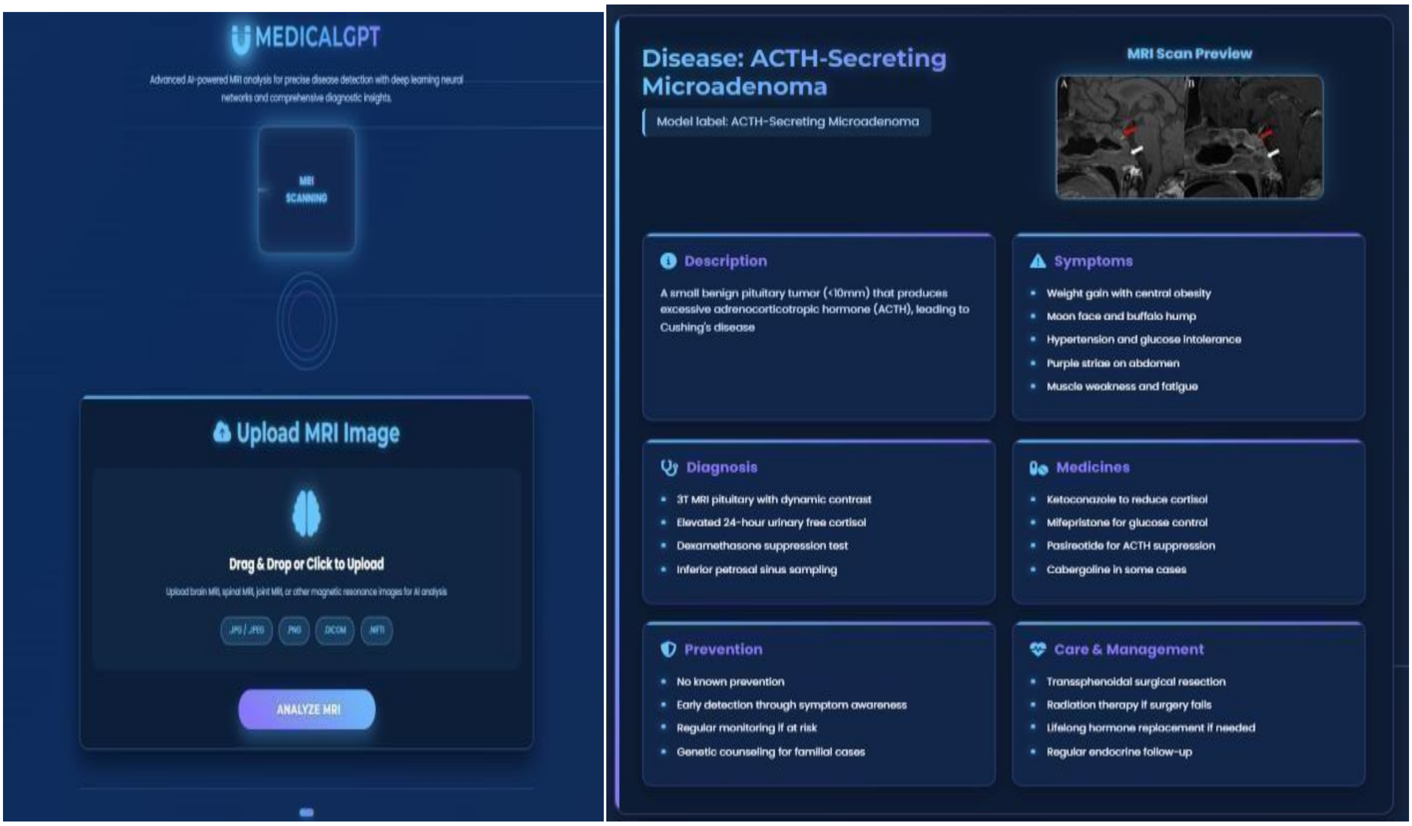
MEDICALGPT Dashboard



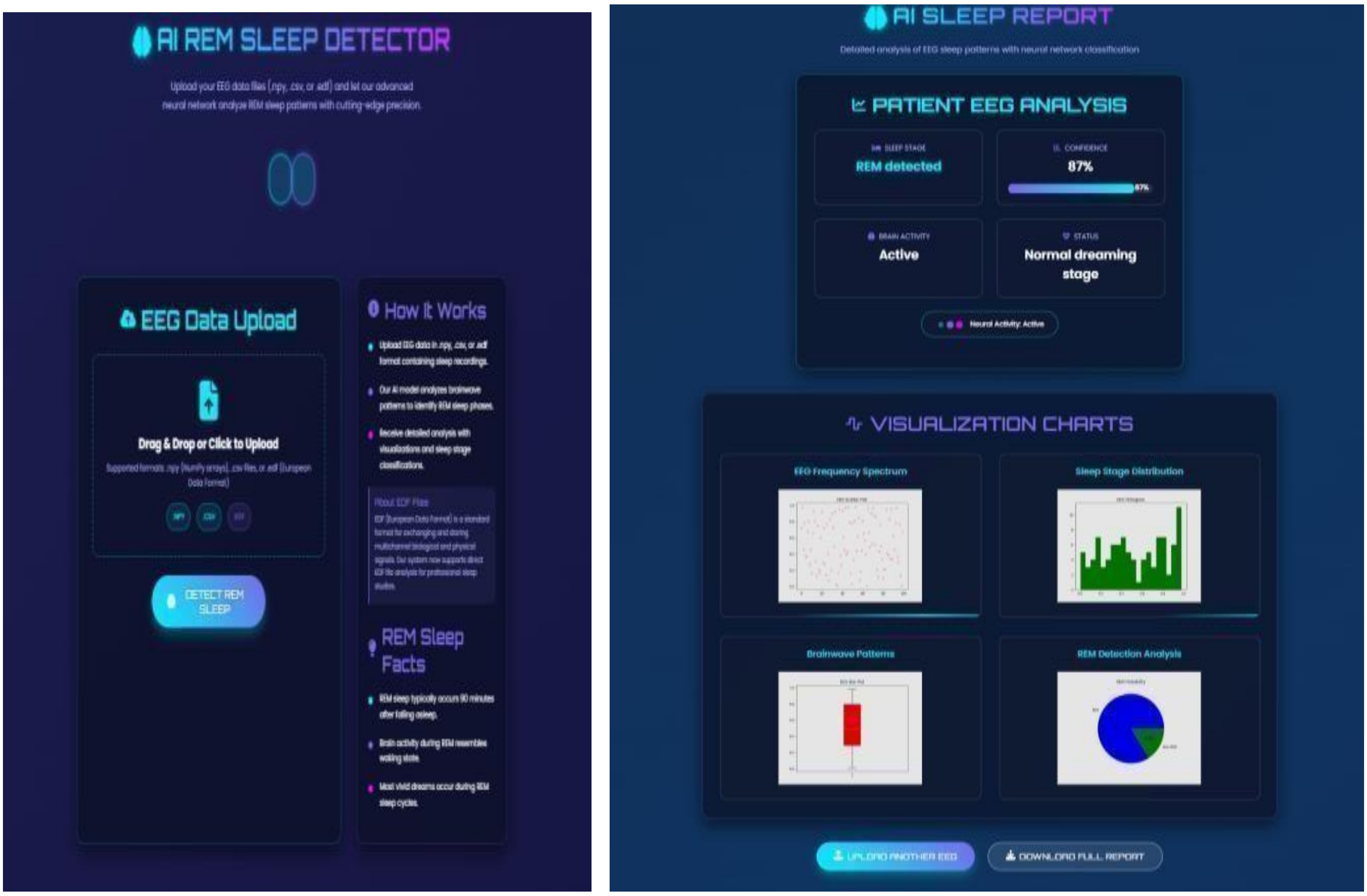
CT Scan Result



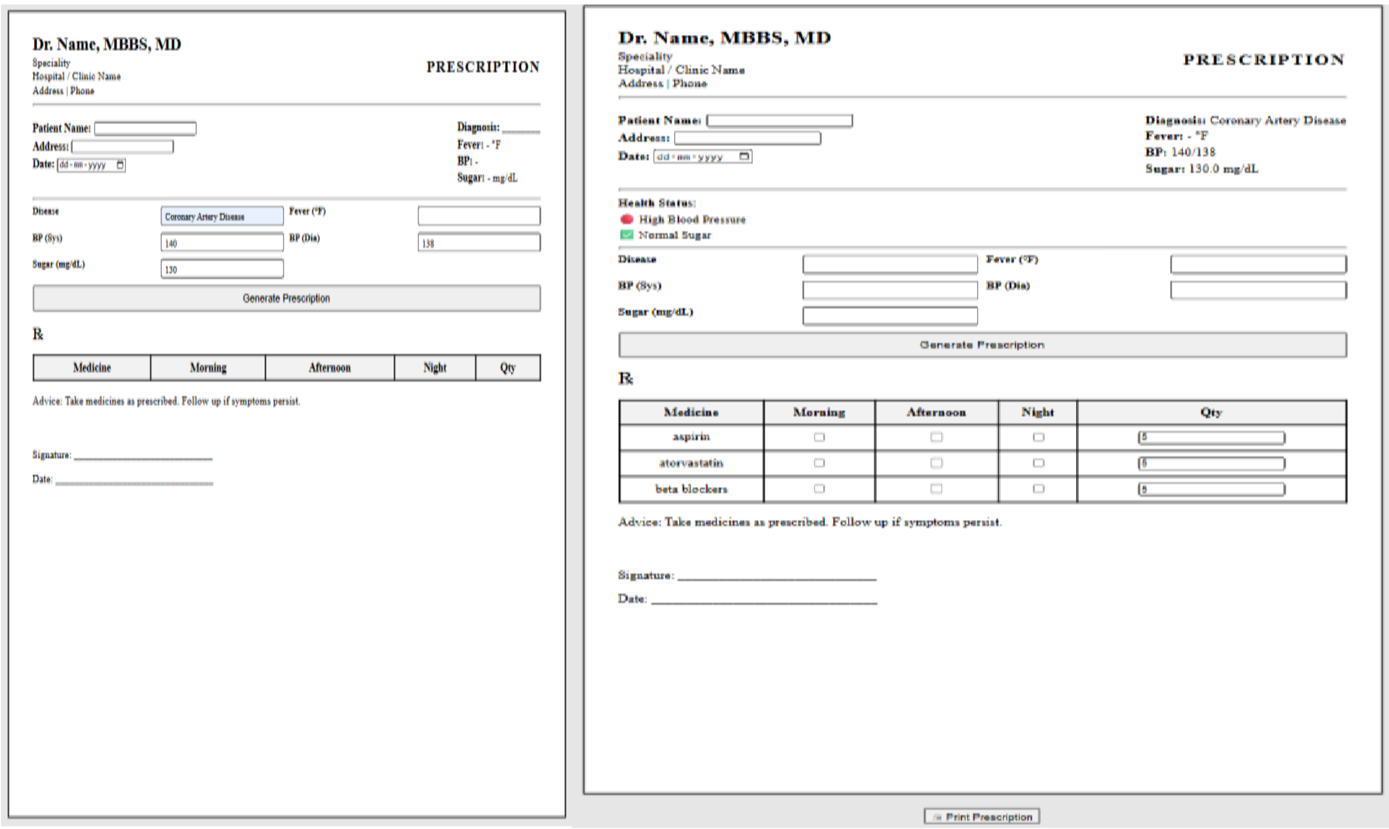
X-RAY Result



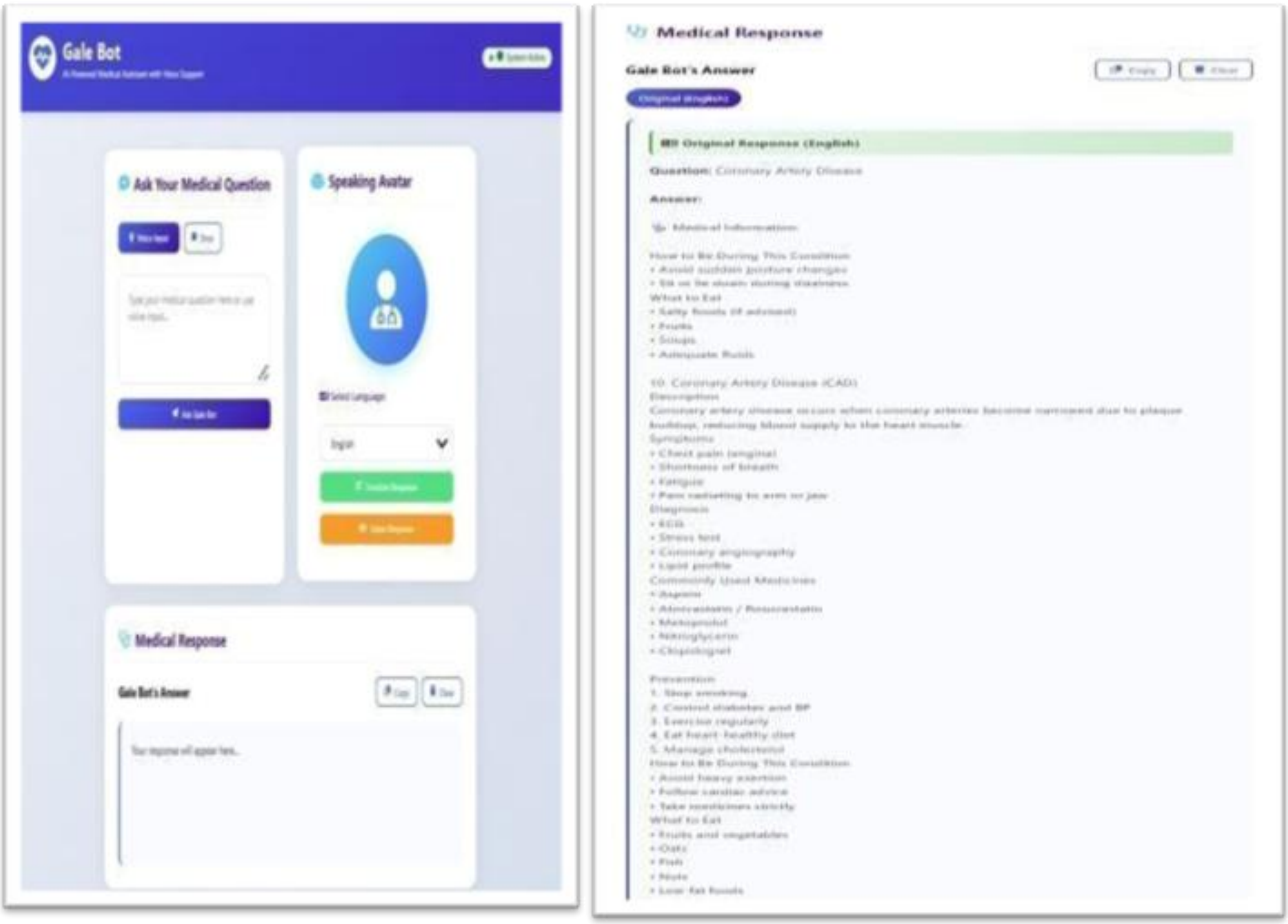
MRI Result



REM Sleep Detector Result



Prescription Generator Result



Medical Bot

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